Identifying treatment effects with a random filter

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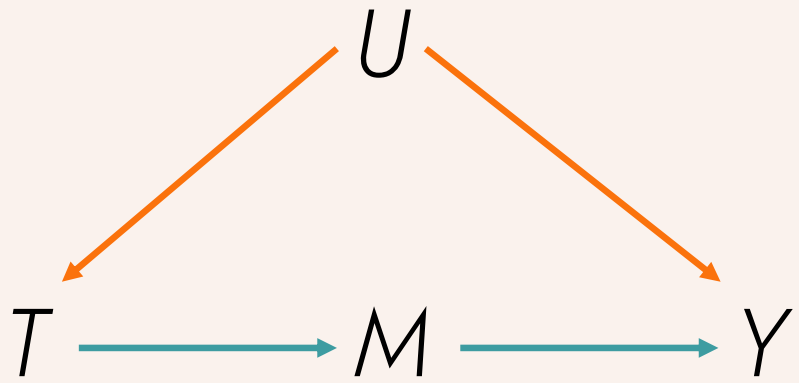
A common challenge of empirical research is estimating the effect of an intervention that permitted selection into treatment.

Goal: Measure the effect of a financial inclusion intervention that provides subsidized mobile banking to El Salvadorans.

Threat: Those who enroll are likely to be better off financially even without this account (prudence, savvy).

Proposal: Experimentally vary bank functionality. This mechanism filters out a selection effect once you condition on enrollment.

We can simulate a dataset with a few lines of code to test this identification strategy. The ATT is a $200 increase in yearly savings.



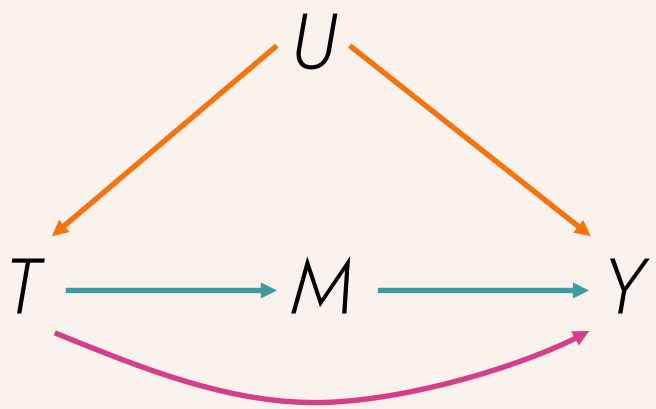
$U \sim Bern(0.5)$	financial savvy
$T \sim Bern(0.25 + 0.5 \cdot U)$	enrollment
$M \sim Bern(0.67 \cdot T)$	functional account
$E \sim N(300, 20^2)$	effect heterogeneity
$Y \sim N(500 + 500 \cdot U + E \cdot M, 25^2)$	yearly savings

The random filter identification strategy removes selection bias by leveraging an exogenous mediator variable (functionality).

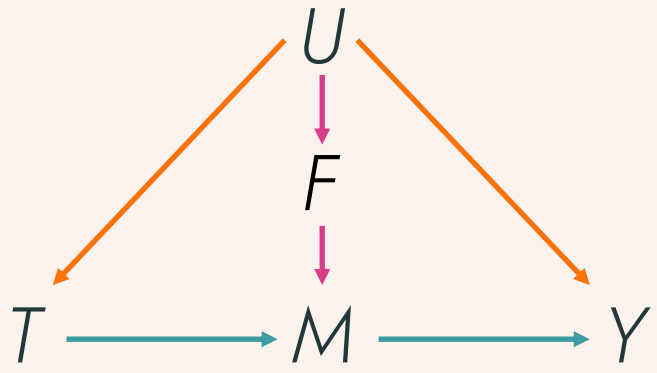
	OLS		Random Filter	
	naïve	benchmark	first	second
	Y	Y	M	Y
T	453.77	**199.74**	0.67	245.67
	(4.81)	**(2.37)**	(0.01)	(6.19)
M	.	.	.	310.54
				(6.57)
U	.	503.83	.	.
		(2.37)		
cons	623.86	498.95	0.00	623.86
	(3.38)	(1.55)	(0.00)	(3.06)

$\beta = \$200$. Random Filter estimate: $0.67 \cdot \$310.54 = \$208.06$.

Aside: I am not assuming the mechanism intercepts the full effect, and my paper extends to conditional exogeneity.



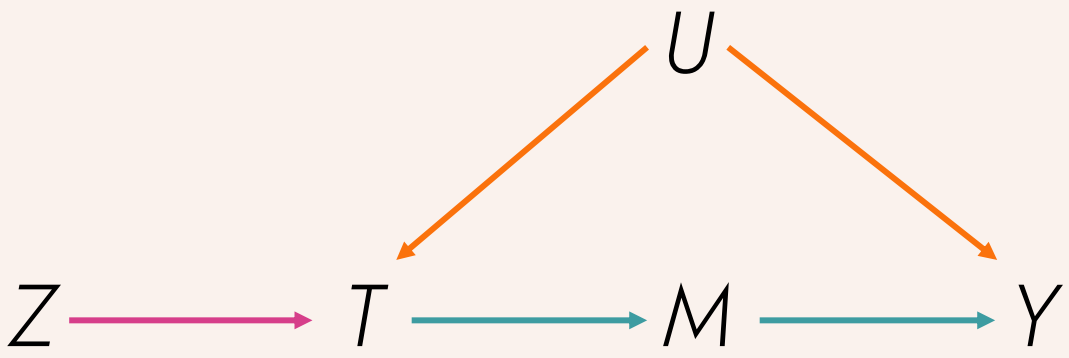
(a) The random filter will only measure the effect through M. Other direct or indirect mechanisms will be ignored.



(b) If M is only conditionally exogenous, controlling all determinants of functionality restores identification.

Ok, exogenous mediator. That's identification. Why is it an ATT?

We can add random assignment to enrollment to determine the difference in instrumental variable and random filter behavior.



$U \sim Bern(0.5)$	financial savvy
$Z \sim Bern(0.5)$	assignment
$C \sim Bern(0.5)$	complier (latent)
$T \sim [C = 0] \cdot Bern(0.25 + 0.5 \cdot U) + [C = 1] \cdot Z$	enrollment
$M \sim Bern(0.67 \cdot T)$	functional account
$E \sim N(450 - 300 \cdot C, 15^2)$	effect heterogeneity
$Y \sim N(500 + 500 \cdot U + E{\cdot}M, 25^2)$	yearly savings

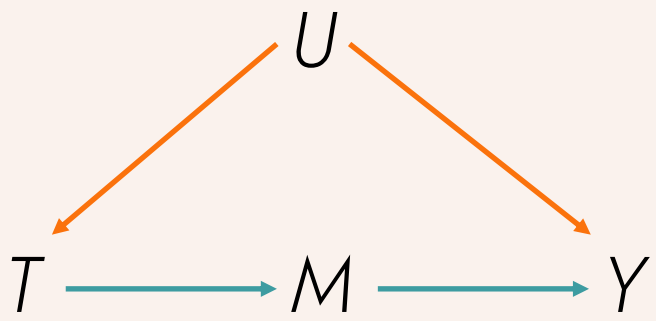
IV weights by compliance and picks up an ATC; RF uses all variation tied to a mechanism and measures an ATT (through M)!

	naïve	benchmark	complier	rf1	rf2	iv1	iv2
	Y	Y	Y	M	Y	T	Y
T	335.04	300.29	99.42	0.66	130.51	.	.
	(5.74)	(3.44)	(1.57)	(0.01)	(7.62)		
M	.	.	.	.	307.65	.	.
					(8.14)		
Z	.	.	.	.	.	0.50	51.21
						(0.01)	(6.63)
U	.	499.20	500.89	.	.	.	.
		(2.45)	(1.56)				
$T \cdot C$	.	-200.84	.	.	.	.	.
		(4.71)					
cons	683.36	494.45	500.14	0.00	683.36	0.25	822.04
	(4.01)	(2.32)	(1.33)	(0.00)	(3.75)	(0.01)	(4.63)

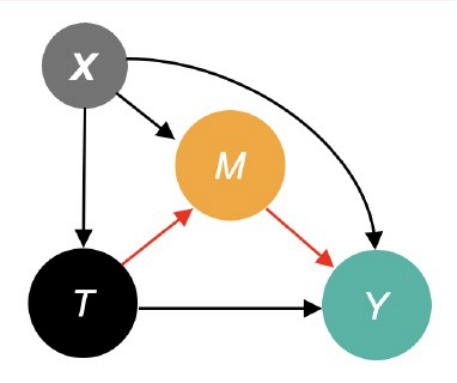
$$\beta_{ATT} = \$200, \ \beta_{ATC} = \$100, \ \hat{\beta}_{RF} = \$203.05, \ \hat{\beta}_{IV} = \$102.42.$$

[Lightning] Contributions

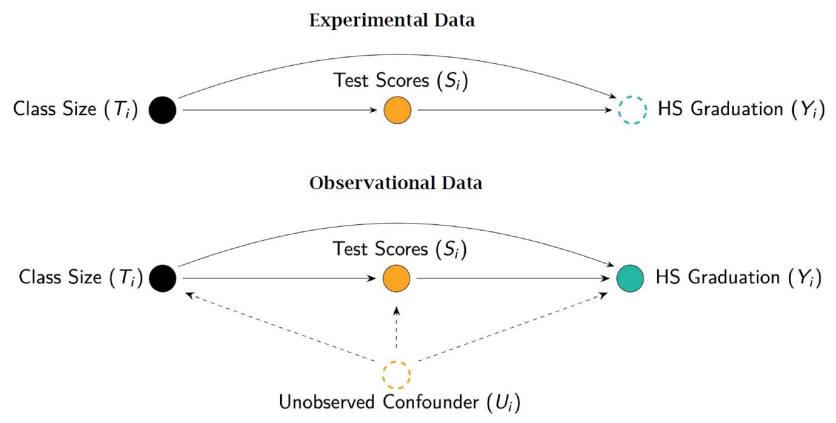
The random filter makes four main contributions.



(a) Expands theory and application of the Front Door Criterion



(b) Relaxes assumptions from Causal Mediation Analysis

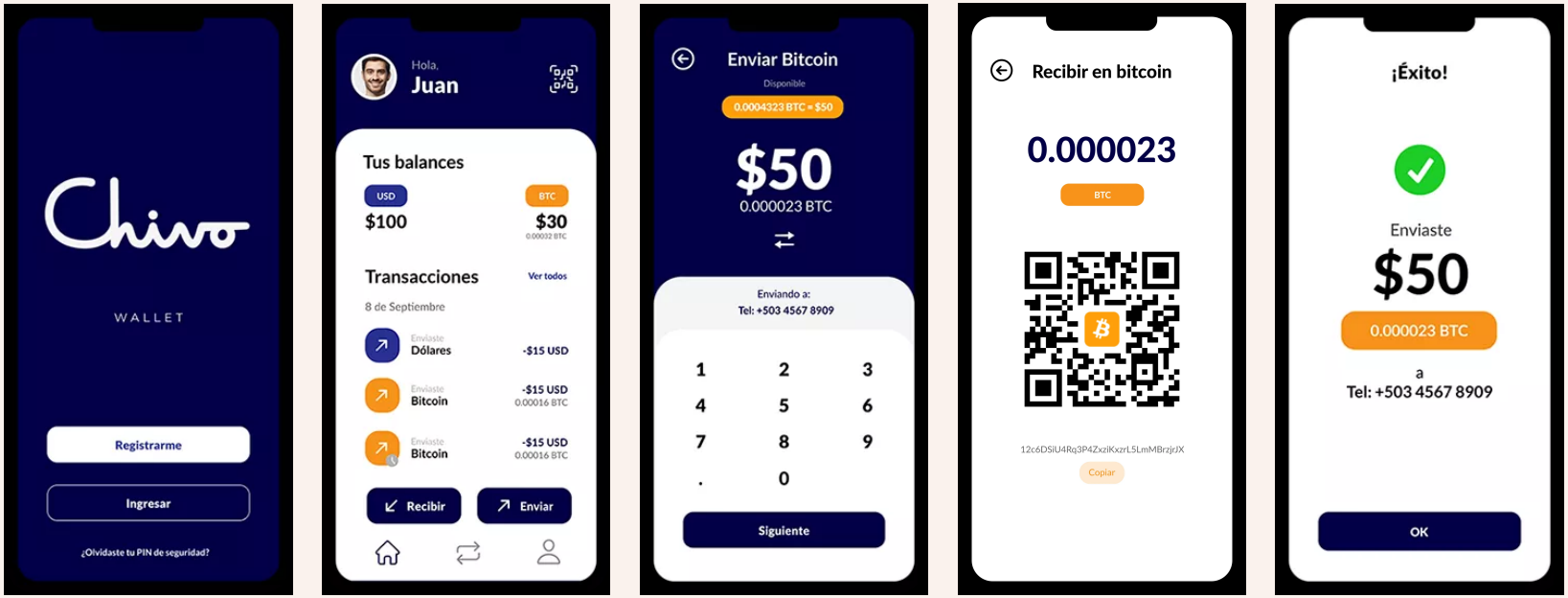


(c) Provides a mediator-driven identification strategy while hot

(d) Shows how an intervention can be hampered by an erosion of trust

Application: Chivo Wallet

Chivo *was* a state-run banking app that touted zero fees on bank transactions, remittances, purchases, and peer-to-peer transfers.



Chivo had potential to benefit El Salvadorans. 70% of adults were unbanked, but 65% had a cell phone.

[†] Chivo, *Chivo App Screenshots* (2021)

Access to Chivo extended to users without a mobile phone, too. Hundreds of staffed ATMs were installed across El Salvador.

[†]Marvin Recinos, *A man in a Nayib Bukele mask next to a Chivo ATM* (2021)

President Bukele further incentivized take-up with three days of wages. 53% of the adult population enrolled in three months.

[†]Press Secretariat of the Republic of El Salvador, *Nayib Bukele announcing a $30 airdrop of BTC to those who sign up for Chivo* (2021)

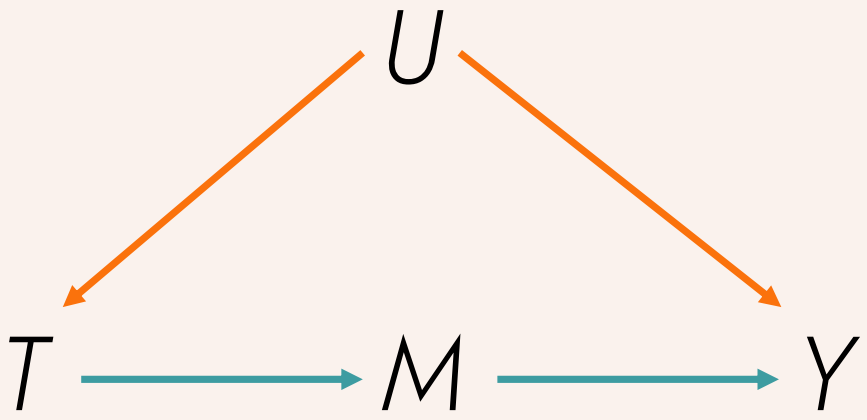
But the platform didn’t work.

Chivo Wallet frequently malfunctioned, leading to inaccessibility beyond users' control and a subsequent loss of public trust.

The government recently abandoned Chivo after years of disuse.

†Ivan Manzano, *Bitcoin ATM kiosk on fire* (2021)

If everyone stopped using Chivo, even those who didn't experience a problem, then the impact of this intervention is zero.



The random filter could verify that the ATT is zero:

Compare financial outcomes (Y) of those who did/did not experience functionality (M), conditional on enrollment (T), and scale this up by the likelihood of access to a functional Chivo.

Survey data supported the hypothesis that any observed difference in finances between enrolled and non-enrolled would be due to selection bias.

The enrolled and non-enrolled are very different. (1/3)

	Enrolled in Chivo		
Variable	$T = 0$	$T = 1$	Difference
no savings	0.347	0.321	-0.026
	(0.477)	(0.467)	(0.041)
money in bank	0.318	0.392	0.075*
	(0.467)	(0.489)	(0.043)
money in house	0.324	0.245	-0.078**
	(0.469)	(0.431)	(0.039)
unbanked	0.512	0.389	-0.123***
	(0.501)	(0.488)	(0.043)
bank account	0.424	0.511	0.088**
	(0.496)	(0.500)	(0.044)
bank trips/month	1.244	1.949	0.705***
	(1.728)	(2.144)	(0.181)
Observations	170	530	700

The enrolled and non-enrolled are very different. (2/3)

	Enrolled in Chivo		
Variable	$T = 0$	$T = 1$	Difference
bank travel time	43.556	36.954	-6.602
	(39.343)	(42.611)	(4.223)
credit card	0.112	0.145	0.034
	(0.316)	(0.353)	(0.030)
remittance change	2.435	2.527	0.092
	(0.866)	(1.107)	(0.142)
cash vs digital	1.854	2.517	0.663***
	(1.235)	(1.476)	(0.132)
only uses cash	0.609	0.404	-0.205***
	(0.490)	(0.491)	(0.046)
dollars vs bitcoin	1.066	1.391	0.325***
	(0.275)	(0.769)	(0.064)
Observations	170	530	700

The enrolled and non-enrolled are very different. (3/3)

	Enrolled in Chivo		
Variable	$T = 0$	$T = 1$	Difference
female	0.524	0.368	-0.156***
	(0.501)	(0.483)	(0.043)
$18 \leq$ age ≤ 39	0.382	0.564	0.182***
	(0.487)	(0.496)	(0.044)
$40 \leq$ age ≤ 62	0.482	0.355	-0.128***
	(0.501)	(0.479)	(0.043)
age ≥ 63	0.135	0.081	-0.054**
	(0.343)	(0.273)	(0.026)
primary school	0.447	0.325	-0.123***
	(0.499)	(0.469)	(0.042)
high school	0.241	0.360	0.119***
	(0.429)	(0.481)	(0.041)
college	0.312	0.315	0.003
	(0.465)	(0.465)	(0.041)
Observations	170	530	700

Those divided by Chivo functionality were more similar. (1/3)

	Experienced no Chivo issue		
Variable	$M = 0$	$M = 1$	Difference
no savings	0.301	0.330	0.029
	(0.460)	(0.471)	(0.044)
money in bank	0.380	0.398	0.017
	(0.487)	(0.490)	(0.046)
money in house	0.276	0.232	-0.044
	(0.448)	(0.422)	(0.041)
unbanked	0.429	0.371	-0.059
	(0.497)	(0.484)	(0.046)
bank account	0.442	0.542	0.101**
	(0.498)	(0.499)	(0.047)
bank trips/month	1.908	1.967	0.059
	(2.390)	(2.029)	(0.202)
Observations	163	367	530

Those divided by Chivo functionality were more similar. (2/3)

	Experienced no Chivo issue		
Variable	$M = 0$	$M = 1$	Difference
bank travel time	39.443	35.941	-3.502
	(48.681)	(39.917)	(4.418)
credit card	0.172	0.134	-0.038
	(0.378)	(0.341)	(0.033)
remittance change	2.506	2.535	0.029
	(1.193)	(1.075)	(0.143)
cash vs digital	2.426	2.557	0.131
	(1.419)	(1.501)	(0.142)
only uses cash	0.419	0.397	-0.022
	(0.495)	(0.490)	(0.047)
dollars vs bitcoin	1.283	1.438	0.155**
	(0.665)	(0.806)	(0.074)
Observations	163	367	530

Those divided by Chivo functionality were more similar. (3/3)

	Experienced no Chivo issue		
Variable	$M = 0$	$M = 1$	Difference
female	0.423	0.343	-0.080*
	(0.496)	(0.475)	(0.045)
$18 \leq$ age ≤ 39	0.528	0.580	0.053
	(0.501)	(0.494)	(0.047)
$40 \leq$ age ≤ 62	0.337	0.362	0.025
	(0.474)	(0.481)	(0.045)
age $\geq$ 63	0.135	0.057	-0.078***
	(0.343)	(0.233)	(0.026)
primary school	0.288	0.341	0.052
	(0.454)	(0.475)	(0.044)
high school	0.387	0.349	-0.038
	(0.488)	(0.477)	(0.045)
college	0.325	0.311	-0.015
	(0.470)	(0.463)	(0.044)
Observations	163	367	530

What did the random filter have to say about the effect of Chivo Wallet?

Downloading a broken app had no effect on financial well-being. The random filter suggests any correlation is due to selection.

	OLS	Random Filter (SUR here)	
	finance index	functional Chivo	finance index
intercept	3.073***	0.000	3.073***
	(0.089)	(0.031)	(0.088)
enrolled Chivo	0.301***	0.691***	0.316***
	(0.102)	(0.036)	(0.126)
functional Chivo	.	.	-0.022
			(0.107)
Observations	690	690	530

RF estimate: **-0.015**. Booted error: **0.075** (rejects naïve value).

[Mini] discussion

The random filter expands our empirical toolkit in three ways.

1. It provides a new identification strategy motif. *We will learn to spot exogenous mechanisms.*
2. It permits new experiments that vary activation of a mechanism when treatment is costly or infeasible to control.
3. It isolates treatment effects through a particular mechanism, which may independently hold policy relevance.

Thank you!

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[†] Marvin Recinos, *Nayib Bukele at a surfing competition* (2021)